

EDUCATION

- **McGill University** Montreal, Canada
Ph.D. in Neuroscience, Supervisor: Prof. Gustavo Turecki Jan 2027 – Expected 2031
- **Wuhan University** Wuhan, China
M.Med. in Psychiatry and Mental Health, Supervisor: Prof. Zhongchun Liu Sep 2023 – Jun 2026
- **Central South University** Changsha, China
B.M. in Clinical Medicine Sep 2018 – Jun 2023

SELECTED PUBLICATIONS

- Zhang L**, Duan H, Lyu H, Ma S, Xu Y, Xie X, Gong Q, Kang L, Chen S, Liu Z. Multi-omics analysis reveals CXCL14+ inhibitory neuron dysfunction in major depressive disorder. *J Affect Disord.* 2026;405:121586. [\[Paper\]](#)
- Zhang L**, Chen X, Xu Y, Liu J, Liu Z. Exploring glucagon-like peptide-1 receptor agonists as potential disease-modifying agent in psychiatric and neurodevelopmental conditions: Evidence from a drug target Mendelian randomization. *BMC Psychiatry.* 2025;25(1):484. [\[Paper\]](#)
- Feng Y, Li N, **Zhang L**, Wang W, Duan H, Sun S, Li R, Zhang Y, Song X, Zhang Y, Zhang H, Nie Z, Yan H, Wang C, Liu Z. X-linked USP11 drives depression-like behaviors by stabilizing CK2 α and disrupting mitochondrial function. *CNS Neurosci Ther.* 2026;32(6):e70934. [\[Paper\]](#)
- Gong Q, Lyu H, Kang L, Ma S, Zhang N, Xie XH, Zhou E, Deng Z, **Zhang L**, Liu J, Liu Z. The genetic relationships between post-traumatic stress disorder and its corresponding neural circuit structures. *J Affect Disord.* 2026;393:120323. [\[Paper\]](#)
- Lyu H, Kang L, Gong Q, Xie XH, Ma S, Yao L, Chen MM, **Zhang L**, Yu H, Wang X, Wang C, Liu Z. The relationship between vitamin D levels and depression: A genetically informed study. *Nutr J.* 2025;24(1):158. [\[Paper\]](#)
- Yuan R, Yang Y, Wang Z, Song Z, **Zhang L**, Guo Z, Yu H, Ju Y, Feng Z. Adolescent psychological resilience and non-suicidal self-injury: A moderated mediation model. *Chin J Clin Psychol.* 2024;32(4):843–849. [\[Paper\]](#)
- Zhang L**, Xu Y, Ma Y, Xie T, Liu C, Liu Q. Research trends in transient receptor potential vanilloid in cardiovascular disease: Bibliometric analysis and visualization. *Front Cardiovasc Med.* 2023;10:1071198. [\[Paper\]](#)

RESEARCH EXPERIENCE

- **A Multidimensional Plasma Proteomic Atlas of Depression** Fudan University
Visiting Student; Supervisor: Prof. Wei Cheng, Institute of Science and Technology for Brain-Inspired Intelligence Sep 2025 – Jan 2026
 - Applied differential association analyses to 7,025 SOMAmer measurements representing 6,365 plasma proteins in the GNPC cohort ($n = 1,346$), identifying 782 proteins associated with depression and validating key markers in the UK Biobank.
 - Integrated proteomic profiles with HADS and MMSE scores and applied non-smooth non-negative matrix factorization, identifying 39 symptom-associated and 11 cognition-associated proteins and resolving three molecularly distinct depression subtypes.
 - Developed a 30-protein proteomic risk score that discriminated patients from controls (AUC = 0.896) and predicted incident depression in Cox regression (HR = 1.145, $p = 1.33 \times 10^{-26}$).
- **Investigating Neuronal Dysfunction in Major Depressive Disorder Through Multi-Omics Integration** Wuhan University
Supervisor: Prof. Zhongchun Liu, Department of Psychiatry, Renmin Hospital of Wuhan University Sep 2023 – Jun 2026
 - Integrated single-nucleus RNA-seq, MDD GWAS and cortical eQTL summary statistics, and UK Biobank plasma proteomics, identifying 273 MDD-associated genes and prioritizing disease-relevant neuronal subtypes.
 - Applied non-negative matrix factorization and trajectory inference to excitatory and inhibitory neurons, revealing disease-associated programs involving synaptic plasticity, neuronal connectivity, and epigenetic regulation.
 - Characterized a *CXCL14*⁺ inhibitory neuron state associated with stress response and cell–cell communication; validated 26 risk genes at the protein level and identified 5 prognostic proteins using CoxBoost regression.
- **Exploring glucagon-like peptide-1 receptor agonists as potential disease-modifying agents in psychiatric and neurodevelopmental conditions** Wuhan University
Supervisor: Prof. Zhongchun Liu, Department of Psychiatry, Renmin Hospital of Wuhan University Sep 2023 – Sep 2024

- Performed drug-target Mendelian randomization by integrating GLP1R cis-eQTL instruments with GWAS data from FinnGen, PGC, and UK Biobank across 12 psychiatric and neurodevelopmental disorders.
- Combined cohort-specific estimates through meta-analysis and assessed robustness using sensitivity analyses, identifying risk-lowering associations for schizophrenia (OR = 0.72), anxiety (OR = 0.87), PTSD, and autism, but an opposing association for OCD.
- **Excellent Doctor Training Program** Central South University
Research Assistant; Supervisor: Prof. Qiming Liu, Department of Cardiology, the Second Xiangya Hospital Mar 2021 – Jun 2023
 - Established ex vivo mouse cardiac perfusion preparations using the Langendorff system and quantified cardiomyocyte gene and protein expression using qPCR and Western blotting.
 - Integrated laboratory training with outpatient and inpatient cardiology rotations and observation of more than ten catheter-ablation procedures, linking experimental measurements to cardiovascular phenotypes and clinical management.
- **Adolescent Psychological Resilience and Non-Suicidal Self-Injury** Central South University
Research Assistant; Supervisor: Prof. Yan Zhang, National Center for Mental Disorders, the Second Xiangya Hospital Mar 2021 – Jun 2023
 - Designed and coordinated a school-based survey of 1,100 adolescents in Changsha, localized validated scales, and produced a quality-controlled analytic sample of 1,011 participants.
 - Used SPSS 26.0 and R for common-method-bias testing, correlation analysis, binary logistic regression, and moderated mediation analysis; estimated an NSSI prevalence of 22.35% and identified depression, anxiety, residence, loneliness, sleep problems, and low resilience as significant correlates.
 - Established a moderated mediation model showing that sleep problems amplified the association between loneliness and NSSI, providing evidence for resilience- and sleep-focused prevention strategies.
- **Exploring the Role and Mechanisms of Wnt1 in Postmenopausal Osteoporosis** Central South University
Supervisor: Prof. Tianlong Huang, the Second Xiangya Hospital Mar 2020 – Mar 2021
 - Established an MC3T3 cell model of postmenopausal osteoporosis and used molecular assays to quantify changes in non-canonical Wnt1 signaling.
 - Confirmed Wnt1 overexpression and generated evidence linking Wnt1 activity to the mTORC1 pathway.

SCHOLARLY CONTRIBUTION

Contributed to the development of *the 2025 Chinese Guidelines for the Diagnosis and Treatment of Bipolar Disorder* under the supervision of Prof. Zhongchun Liu. Responsible for drafting the background section, conducting literature reviews, integrating clinical evidence, and formulating treatment recommendations. Acknowledged for contributions in the final published guideline.

CONFERENCES & PRESENTATIONS

- **The 22nd Annual Conference of the Chinese Psychiatrist Association** Tianjin, China
Poster Presentation July 2026
 - **Title:** *Mapping Transdiagnostic Symptom Dimensions to Brain Cell-State Programs in Major Psychiatric Disorders*

HONORS & AWARDS

McGill-CSC Joint Scholarship	China Scholarship Council, 2026
Outstanding Graduate Student Award	Wuhan University, 2025
First Prize Scholarship	Wuhan University, 2025
Outstanding Social Practice Team Award	Central South University, 2021
Outstanding Student Award	Central South University, 2019
First Prize Scholarship	Central South University, 2019
Zhangyuan Tungsten Industry Scholarship	Hongyi Zhangyuan Tungsten Co., Ltd & Central South University, 2019

SPECIALIZED SKILLS & INTERESTS

- **Programming & Data Visualization:** R, Python, bash, LaTeX; PyTorch, TensorFlow, scikit-learn, pandas, NumPy, ggplot2, Matplotlib, Seaborn.
- **Bioinformatics Tools & Pipelines:** Seurat, Scanpy, ArchR, cNMF, Cell Ranger, Signac, PLINK, Snakemake.
- **Experimental Techniques:** Western blotting, qPCR, immunofluorescence, cell culture, animal handling.
- **Languages & Research Interests:** Mandarin (native); English (IELTS Academic: Overall Band 6.5); computational psychiatry, psychiatric genomics, single-cell and spatial omics, and multi-omics integration.